

Resource and Energy Sustainability: Vietnam vs. 7 Southeast Asian Countries

Course: Data Visualization — International University, VNU-HCM (HCMIU)

Repository: <https://github.com/LeXuanQue/sea-resource-sustainability-viz>

Member	Student ID	Responsibility
Trần Nguyễn Lê Quân (leader)	ITDSIU25033	Data: processing, normalising, JSON export
Huỳnh Ngọc Trúc Phương	ITDSIU25031	UI/UX: interface design, chart selection, colours
Lê Xuân Quế	ITDSIU25036	Frontend: HTML/CSS skeleton and main charts
Đặng Hoàng Quân	ITDSIU25035	Frontend: JavaScript interactions and layer transitions

1. Background and motivation

Natural resources and energy are closely tied to the long-term sustainability of economic development. The World Bank's Sovereign ESG framework evaluates the environmental dimension of economic performance through a country's natural-resource endowment and management alongside sustainable energy use, which makes forest resources, resource depletion, freshwater use, energy intensity, renewable energy and fossil-fuel dependence relevant dimensions of environmental sustainability. For Viet Nam the topic is also directly relevant to national policy: the National Climate Change Strategy targets net-zero greenhouse-gas emissions by 2050, identifies resource management, forest protection, energy and land-use change as priority areas, and sets a 2050 target of maintaining forest cover at 43%. The World Bank records Viet Nam's forest area at 47.2% of land area in 2022; the two figures use different forest definitions and are not directly comparable.

The energy dimension matters because Viet Nam is seeing strong growth in electricity demand while expanding renewable generation. According to Ember, Viet Nam's electricity demand grew by an average of 8.2% per year between 2013 and 2024, and coal-based generation expanded over the same period to meet that rising demand; at the same time, Viet Nam recorded substantial growth in wind and solar generation. This reflects a wider regional challenge: Ember reported that coal supplied 45% of ASEAN electricity generation in 2024 while regional demand kept growing. Examining Viet Nam alone cannot show its position in that context, whereas comparing it with Indonesia, Thailand, Malaysia, the Philippines, Cambodia, Laos and Myanmar reveals differences in resource conditions, resource pressure and energy transition without assuming one indicator represents sustainability as a whole.

A static report suits this poorly, because 12 indicators across 8 countries and many years form a multidimensional dataset needing repeated comparison across indicators, countries and time. An interactive interface lets users pick an indicator, year or country and read the trend and regional comparison directly. The aim is not a single composite sustainability score, but to help viewers explore where Viet Nam stands, how its indicators have moved, and how those patterns compare with its neighbours.

2. Objectives

This project aims to develop an interactive web-based visualization for exploring natural-resource and energy sustainability indicators across Viet Nam and seven selected Southeast Asian countries. The visualization will enable users to examine the historical development of 12 indicators, compare Viet Nam with the selected regional countries, and observe how Viet Nam's relative position changes over time. The project focuses on supporting data exploration and comparison rather than producing a single composite sustainability score.

Specifically, the interactive visualization is designed to answer the following research questions:

- **RQ1:** How have Viet Nam's 12 natural-resource and energy indicators changed over time?
- **RQ2:** How does Viet Nam compare with Indonesia, Thailand, Malaysia, the Philippines, Cambodia, Laos and Myanmar across the 12 indicators over time?
- **RQ3:** How does the gap between Viet Nam and the selected-country regional median change over time for the 12 indicators?

3. Data: where and how we collect it

All data comes from one public source: the World Bank Sovereign ESG dataset, downloaded once as a CSV (`WB_ESG_WIDEF.csv`, 239 economies, 71 core indicators, 1960–2023), so no scraping or API is needed and the dataset is fixed and reproducible. From it we keep 8

countries — Vietnam, Indonesia, Thailand, Malaysia, the Philippines, Cambodia, Laos and Myanmar — and 12 indicators from two Environment themes, all present for all 8 countries with coverage checked row by row.

Natural resources (code)	Unit, years	Energy (code)	Unit, years
Forest area (AG.LND.FRST.ZS)	% of land area, 1990–2022	Energy intensity (EG.EGY.PRIM.PP.KD)	MJ / \$2017 PPP GDP, 2000–2022
Tree cover loss (AG.LND.FRLS.HA)	hectares, 2002–2021	Renewable energy consumption (EG.FEC.RNEW.ZS)	% of final energy, 1990–2022
Protected areas, land and marine (ER.PTD.TOTL.ZS)	% of territory, 2013–2023	Renewable electricity output (EG.ELC.RNEW.ZS)	% of electricity, 1990–2021
Natural resources depletion (NY.ADJ.DRES.GN.ZS)	% of GNI, 1990–2021	Energy use per person (EG.USE.PCAP.KG.OE)	kg oil equivalent, 1990–2022
Net forest depletion (NY.ADJ.DFOR.GN.ZS)	% of GNI, 1990–2021	Fossil fuel consumption (EG.USE.COMM.FO.ZS)	% of total energy, 1990–2022
Freshwater withdrawals (ER.H2O.FWTL.ZS)	% of internal res., 1990–2021	Electricity from coal (EG.ELC.COAL.ZS)	% of electricity, 1990–2022

Coverage is uneven for four countries: Cambodia starts in 1995 on most series and Laos in 2000 on energy use, fossil fuel and coal; freshwater withdrawals start in 2005–2007 for Laos, the Philippines, Cambodia and Thailand; and Laos reports renewable electricity for only 10 scattered years between 2001 and 2021. Source: World Bank Sovereign ESG Data Portal.

4. Data processing

A short Python script turns the wide CSV into three static JSON files the site reads directly; nothing is computed in the browser except filtering. **(1) Filter** to the 8 countries, the 12 indicators above and years 1990–2023. **(2) Reshape** from wide form to long form: one row per country, indicator, year and value. **(3) Keep gaps as gaps** — a missing value is stored as null, never 0, and charts never interpolate across it; a late-starting series gets a hollow start marker and an isolated year a single dot. **(4) Flag repeated values** — some series repeat one estimate for years, as Vietnam's freshwater withdrawals sit at exactly 22.78% from 2005 to 2021; these runs are drawn lighter so they are not read as new measurements. **(5) Choose the year shown** — each indicator defaults to its latest year with at least 6 of the 8 countries reporting (2021, 2022 or 2023), printed beside every value. **(6) Regional reference = median** — with 8 countries of very different size one can drag the mean far (freshwater: mean 11.0%, median 7.4%), so the median of those reporting that year is used everywhere; on trend charts it uses only countries with continuous data in the window shown, so it does not jump when a country enters. **(7) Direction and rank** — each indicator is tagged "higher is better" or "lower is better" and ranks are direction-adjusted, so rank 1 is always the best outcome; energy use per person has no agreed direction, so it gets no verdict and no rank. **(8) Verdict** — Vietnam is "better" or "worse" by its position against the median and "level" within 3% of it; change since 2010 is judged separately, because the two can disagree. **(9) Export** values.json (long form), snapshot.json (latest value, median and rank per indicator) and coverage.json (first year, last year and gaps per country).

Known limitations. Tree cover loss is in hectares, so larger countries look worse by construction; protected areas has no 2010 baseline; all values are national averages.

5. Visualization design: three alternative prototypes

All three designs answer the same question with the same three levels — region, one country, one indicator over time. They differ in how the reader moves between those levels. Each is drawn in the project Figma file and exported to docs/sketch/sketch-1.png ... sketch-6.png, two pages per design.

Design A: story first, then explore (Trần Nguyễn Lê Quân)

The page opens as a short explanation and only then hands control to the reader, following the "martini glass" structure of narrative visualization (Segel and Heer, 2010).

- **Hero.** The question as the title, a one-sentence answer, and three headline numbers with a plain "better / worse than region" tag: forest area +18.4 points since 1990, renewable energy share –51.7 points since 1990, and 2.9% of territory protected.
- **Three story blocks.** Each has a conclusion as its title, one simple chart and two sentences: Vietnam is the only country whose forests grew back substantially; it gave up its renewable energy share faster than any neighbour; and it protects less of its territory than any of them.
- **Explore section.** One shared indicator selector, country selector and year slider drive three linked panels: a ranked bar list with an orientation map (region level), a dot plot of all

12 indicators for the selected country (country level), and a trend line against the regional median (indicator level). Clicking a country in the list or map drills down to that country; a "Back to Vietnam" button returns.

- **Details.** A collapsed section with the full 12-indicator table, data coverage per country and the method notes.

Only Vietnam is coloured; the other seven countries are grey and named on hover. Better and worse are shown with symbol, word and colour together (▲ better in teal, ▼ worse in orange), so the page reads correctly for colour-blind viewers.

Strengths: a reader who never touches a control still gets the main message; every chart uses position or length, the encodings people read most accurately. **Weaknesses:** the page is long, and the story blocks must be rewritten if the data changes.

Design B: coordinated dashboard (Trần Nguyễn Lê Quân)

Everything sits on one screen with no scrolling story. A top bar holds the three shared controls — indicator, country and year — and every panel below answers to them at once.

- **KPI strip.** Six cells under the tabs "Natural resources | Energy", each giving the value, its unit, the year it comes from, and a ▲ better / ▼ worse verdict against the regional median.
- **Ranked bar list.** All 8 countries for the chosen indicator and year, best at top, with the regional median drawn as a dashed line across the bars.
- **Small multiples.** Six mini line charts, one per indicator in the active tab, each carrying its own verdict.
- **One large trend chart.** The selected country against the regional median, 1990–2022.

Clicking a country in the ranking re-selects it and the KPI strip, the small multiples and the trend chart all switch together; a removable chip shows the current selection. Vietnam keeps its own colour in every view, so it stays findable even while another country is selected.

Strengths: every level is visible at the same time, so comparing one indicator against another costs no navigation, and no state is hidden. **Weaknesses:** the main message is not readable without interacting, and six panels at once is dense for a first-time reader.

Design C: head-to-head comparison (Đặng Hoàng Quân)

Layout. Three full-page levels inside one "Sustainability Atlas" shell with a fixed 01 / 02 / 03 header. Level 1, Regional Overview, pairs a Southeast Asia choropleth with a ranked bar list of the 8 countries and cards for the regional average and median. Level 2, Country Dashboard, shows all 12 indicators as sparkline cards for one country against a chosen comparison country. Level 3, Indicator Trend, is a single multi-line time-series chart with country chips, a draggable year cursor and a per-country coverage table.

Interaction. The reader moves down the levels along a visible path: clicking a country on the map or in the ranking opens Level 2, clicking any indicator card opens Level 3, and a numbered breadcrumb (01 → 02 → 03, with "You are here") plus a back button on every level returns upward. Indicator, country, comparison-country and year controls sit at the top of each level, and "continue the analysis" cards at the foot of Level 1 name the next step explicitly.

Strength. The path between the three levels is the most explicit of the three designs — numbered steps, breadcrumbs, back buttons and inline interaction hints mean the reader never loses their place — and missing data is handled rigorously throughout: grey hatching on the map, labelled "no data" bands on the trend chart, named gaps for Cambodia before 2005 and Laos before 2002, and "gaps are unavailable observations, never zero" stated on every level.

Weakness. It benchmarks Viet Nam against the regional *average* by default, through a "Gap vs regional average" card and an "Average + median" benchmark selector, which conflicts with our rule that the regional reference is the median; Level 2 compares Viet Nam against a single partner country rather than against the region; and three full-length pages is the largest build of the three.

Nothing is missing from the required checklist: all three levels are present with a visible path between them, indicator, country and year controls appear on every level, missing data is shown explicitly rather than as zero, and units and the World Bank Sovereign ESG source are printed on each page.

Chosen direction

Criterion	Design A	Design B	Design C
Main message readable without interaction	Yes	No — needs a selection first	Partly — Level 1 states one finding
All three levels and the path between them	Yes, drill-down and back	Yes, all three on one screen	Yes, the most explicit path
Shows missing data explicitly	Yes	Yes	Yes
Effort to build in 10 weeks with no prior frontend experience	Medium	Medium to high	High — three full pages

We will implement Design A. It is the only one of the three whose main finding is readable with no interaction, and it still contains the full explore section, so it answers every research question while staying the lowest-risk build for a team with no prior frontend experience.

6. Must-have and optional features

Must-have features are what the week-6 prototype needs to answer the research questions; optional features follow only if time allows after week 7.

Must-have. M1 Region level — ranked bar list of the 8 countries for one indicator and year, regional median marked. **M2 Country level** — all 12 indicators for one country, each with its rank among the 8 and its verdict against the median. **M3 Indicator level** — trend line from 1990, selected country highlighted, regional median dashed. **M4 Shared controls** — indicator selector, country selector and year slider, every panel updating together. **M5 Drill-down** — clicking a country in the ranking opens its country view; a back control returns. **M6 Tooltips** — country, year, value and unit on every mark. **M7 Missing data** — shown explicitly ("no data", line breaks, hatching), never as zero. **M8 Provenance** — units on every chart and the World Bank ESG source on the page. **M9 Delivery** — static site on GitHub Pages, no backend. **M10 Story section** — hero with three headline findings and three story blocks, readable without interaction.

Every research question is answered by at least one must-have: **RQ1** by M3, M2 and M10, **RQ2** by M1, M4 and M5, **RQ3** by M3, M2 and M10.

Optional. Orientation map beside the ranking; collapsible details section with the full indicator table and per-country coverage; animated transitions on year or indicator change; shareable links that keep the selection in the URL; Vietnamese and English interface versions.

7. Project schedule

Ten weeks from Monday 5 October 2026; the full plan with per-member tasks is in the repository at docs/project-schedule.md.

Wk	Milestone (week beginning)	Wk	Milestone (week beginning)
1	Oct 5 — kickoff: repo, proposal, sketches	6	Nov 9 — working prototype due , on GitHub Pages
2	Oct 12 — choose design; JSON schema; page skeleton	7	Nov 16 — feedback and improvements
3	Oct 19 — JSON v1; region chart; shared controls	8	Nov 23 — optional features; report sections
4	Oct 26 — country and indicator levels; drill-down	9	Nov 30 — feature freeze, testing, report draft
5	Nov 2 — midterm: integration and bug fixes	10	Dec 7 — final submission

Weeks 3–4 are led by Trần Nguyễn Lê Quân (data) with Lê Xuân Quế and Đặng Hoàng Quân (frontend); all other weeks are shared. Weeks 1–2 include time to learn D3.js, as no member has built a frontend before.

8. Sources

World Bank — Sovereign ESG Data Framework: <https://esgdata.worldbank.org/data/framework>. World Bank — About the Sovereign ESG Data Portal: <https://esgdata.worldbank.org/about/aboutus?lang=en>. Chính phủ Việt Nam — Chiến lược quốc gia về biến đổi khí hậu đến năm 2050 (National Climate Change Strategy to 2050): <https://xaydungchinhhsach.chinhphu.vn/phe-duyet-chien-luoc-quoc-gia-ve-bien-doi-khi-hau-den-nam-2050-119220727070101019.htm>. Ember — ASEAN electricity data and regional transition: <https://ember-energy.org/data/electricity-data-explorer/>. Ember — From emission-intensive to investment hotspots: championing renewables in 3 ASEAN economies: <https://ember-energy.org/latest-insights/from-emission-intensive-to-investment-hotspots-championing-renewables-in-3-asean-economies/country-snapshots-and-opportunities/>. Segel, E. and Heer, J. (2010). Narrative Visualization: Telling Stories with Data. *IEEE Transactions on Visualization and Computer Graphics*, 16(6).

How sustainable is Vietnam's use of resources and energy, compared with Southeast Asia?

Vietnam has rebuilt its forests faster than any of its neighbours, given up more of its renewable energy than any of them, and protects less of its territory than all of them.

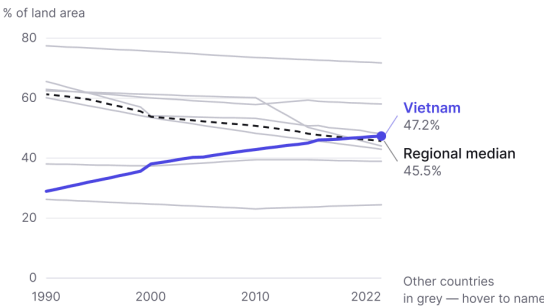
▲ better vs region

+18.4 pts

Forest area gained, 1990–2022

From 28.8% of its land area to 47.2%. Only Thailand also gained, by 0.9 points. Every other country fell.

Three things explain that answer.



Vietnam is the only country whose forests grew back substantially.

Forest cover fell across most of Southeast Asia between 1990 and 2022. Vietnam went the other way — 28.8% of its land area to 47.2%, a gain of 18.4 points — after a national replanting programme begun in the 1990s.

Thailand edged up by 0.9 points. Every other country here lost forest. The regional median in 2022 is 45.6%.

▼ worse vs region

–51.7 pts

Renewable energy share lost, 1990–2021

From 75.9% to 24.2% — the largest fall of the eight. Fifth of eight in 2021; regional median 26.1%.

▼ worse vs region

2.9%

Territory protected, 2023

The lowest of the eight countries. The regional median is 7.1%; Cambodia protects 31.8%.

Sketch 1 — Design A: story first. The main finding is readable without any interaction.

Click a country (drill-down)

Clicking Laos switches every panel to Laos. Vietnam stays on screen as a reference. Renewable electricity also shows the gap rule: Laos reports ten scattered years, and the isolated 2012 reading is a single dot.

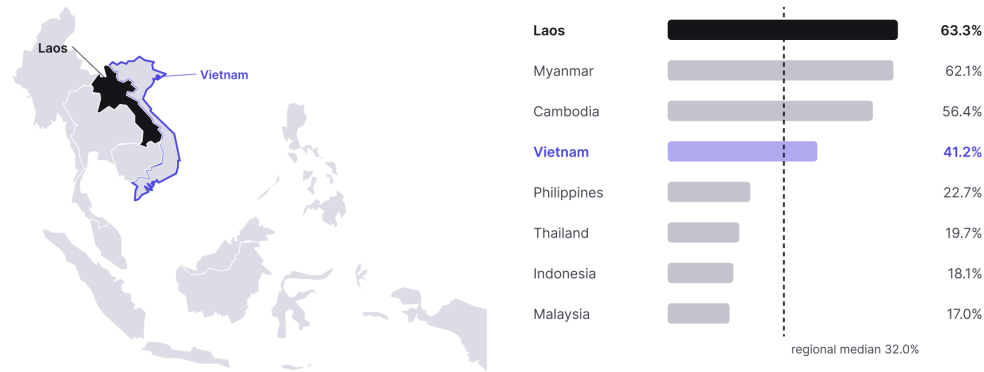
Now explore the rest yourself.

One indicator, one country, one year — shared by all three panels. The year starts at the latest the indicator reports.

Indicator **Renewable electricity** Country **Laos** Year **2021** 1990 2021 [← Back to Vietnam](#)

Laos gets the largest share of its electricity from renewables.

% of electricity, 2021 · best at top · dashed line is the median of the 8 countries reporting



The map is here for orientation — the values are in the list.

↓ click a country

Drag the year to 1995

Every panel follows the year. In 1995 four countries have no freshwater reading, and three whole indicators do not start until later.

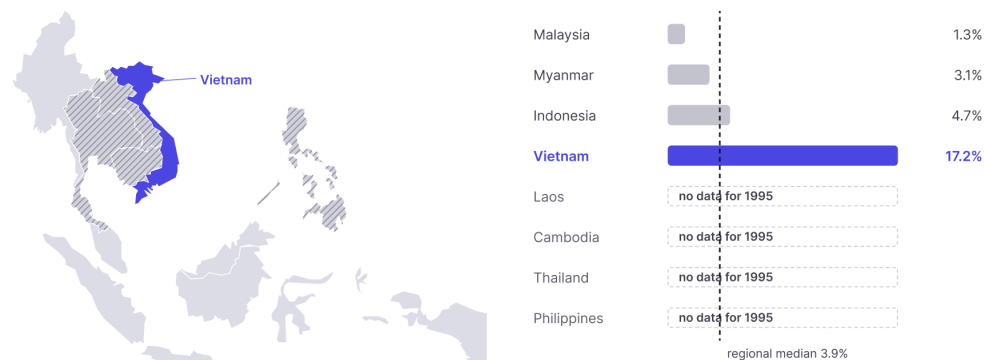
Now explore the rest yourself.

One indicator, one country, one year — shared by all three panels. The year starts at the latest the indicator reports.

Indicator **Freshwater withdrawals** Country **Vietnam** Year **1995** 1990 2021

Only four of the eight countries reported freshwater withdrawals in 1995.

% of internal resources, 1995 · best at top · dashed line is the median of the 4 countries reporting



The map is here for orientation — the values are in the list.

SEA Sustainability

Vietnam vs 7 Southeast Asian countries · World Bank Sovereign ESG

INDICATOR

Forest area (% of land area)

COUNTRY

Vietnam

YEAR

1990

2022

Natural resources

Energy

vs regional median · latest year per indicator

Forest area

47.2

% of land area

▲ better

median 45.6 · 2022

Tree cover loss

193k

hectares

▲ better

median 235k · 2021

Protected areas

2.9

% of territory

▼ worse

median 7.05 · 2023

Resource depletion

0.51

% of GNI

▲ better

median 2.31 · 2021

Net forest depl.

0

% of GNI

= level

median 0 · 2021

Freshwater use

22.8

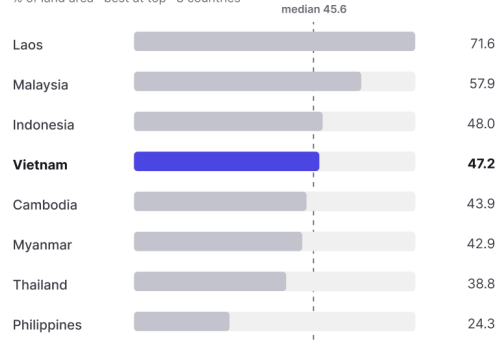
% of internal resources

▼ worse

median 7.44 · 2021

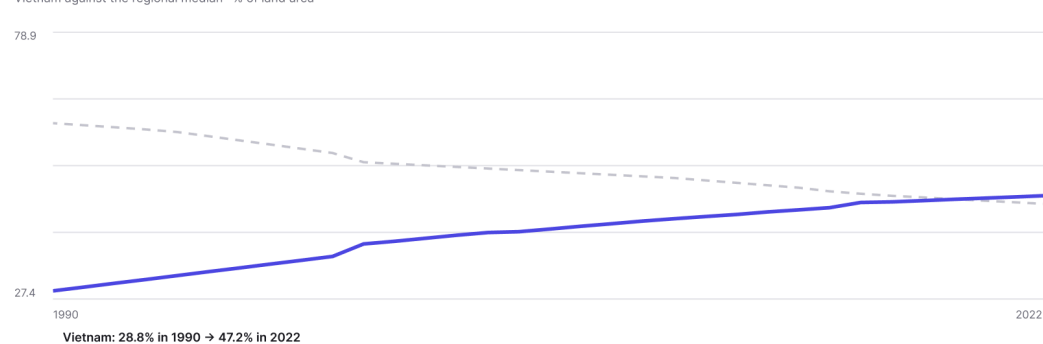
Forest area, 2022

% of land area · best at top · 8 countries



Forest area over time, 1990-2022

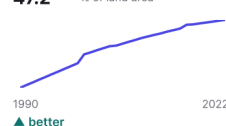
Vietnam against the regional median · % of land area



ALL SIX NATURAL-RESOURCE INDICATORS FOR VIETNAM

Forest area

47.2 % of land area



Tree cover loss

193k hectares



Protected areas

2.9 % of territory



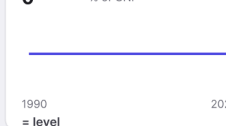
Resource depletion

0.51 % of GNI



Net forest depl.

0 % of GNI



Freshwater use

22.8 % of internal resources



Regional reference is the MEDIAN of the 8 countries, never the mean. Missing values are shown as "no data", never 0. Ranks are direction-adjusted, so rank 1 is always the best outcome.

Source: World Bank Sovereign ESG · esgdata.worldbank.org

Sketch 3 — Design B: coordinated dashboard, every view on one screen.

SEA Sustainability

Vietnam vs 7 Southeast Asian countries · World Bank Sovereign ESG

INDICATOR

Forest area (% of land area)

COUNTRY

Laos

YEAR

1990

2022

Natural resources

Energy

vs regional median · latest year per indicator

Laos selected x

Forest area

71.6

% of land area

▲ better

median 45.6 · 2022

Tree cover loss

318k

hectares

▼ worse

median 235k · 2021

Protected areas

19.0

% of territory

▲ better

median 7.05 · 2023

Resource depletion

3.44

% of GNI

▼ worse

median 2.31 · 2021

Net forest depl.

1.59

% of GNI

▼ worse

median 0 · 2021

Freshwater use

3.86

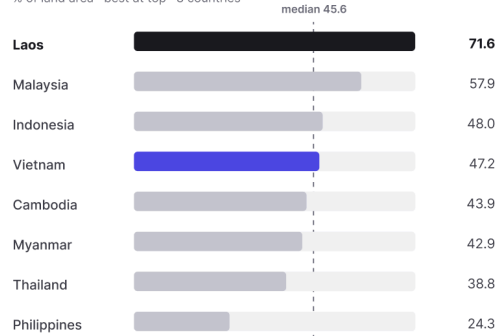
% of internal resources

▲ better

median 7.44 · 2021

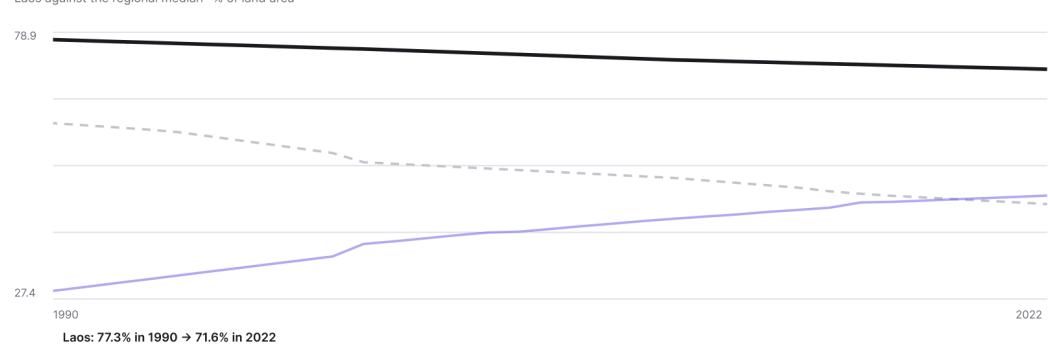
Forest area, 2022

% of land area · best at top · 8 countries



Forest area over time, 1990-2022

Laos against the regional median · % of land area



ALL SIX NATURAL-RESOURCE INDICATORS FOR LAOS

Forest area

71.6

% of land area



1990

2022

▲ better

Tree cover loss

318k

hectares



2002

2021

▼ worse

Protected areas

19.0

% of territory



2013

2023

▲ better

Resource depletion

3.44

% of GNI



1990

2021

▼ worse

Net forest depl.

1.59

% of GNI



1990

2021

▼ worse

Freshwater use

3.86

% of internal resources



2005

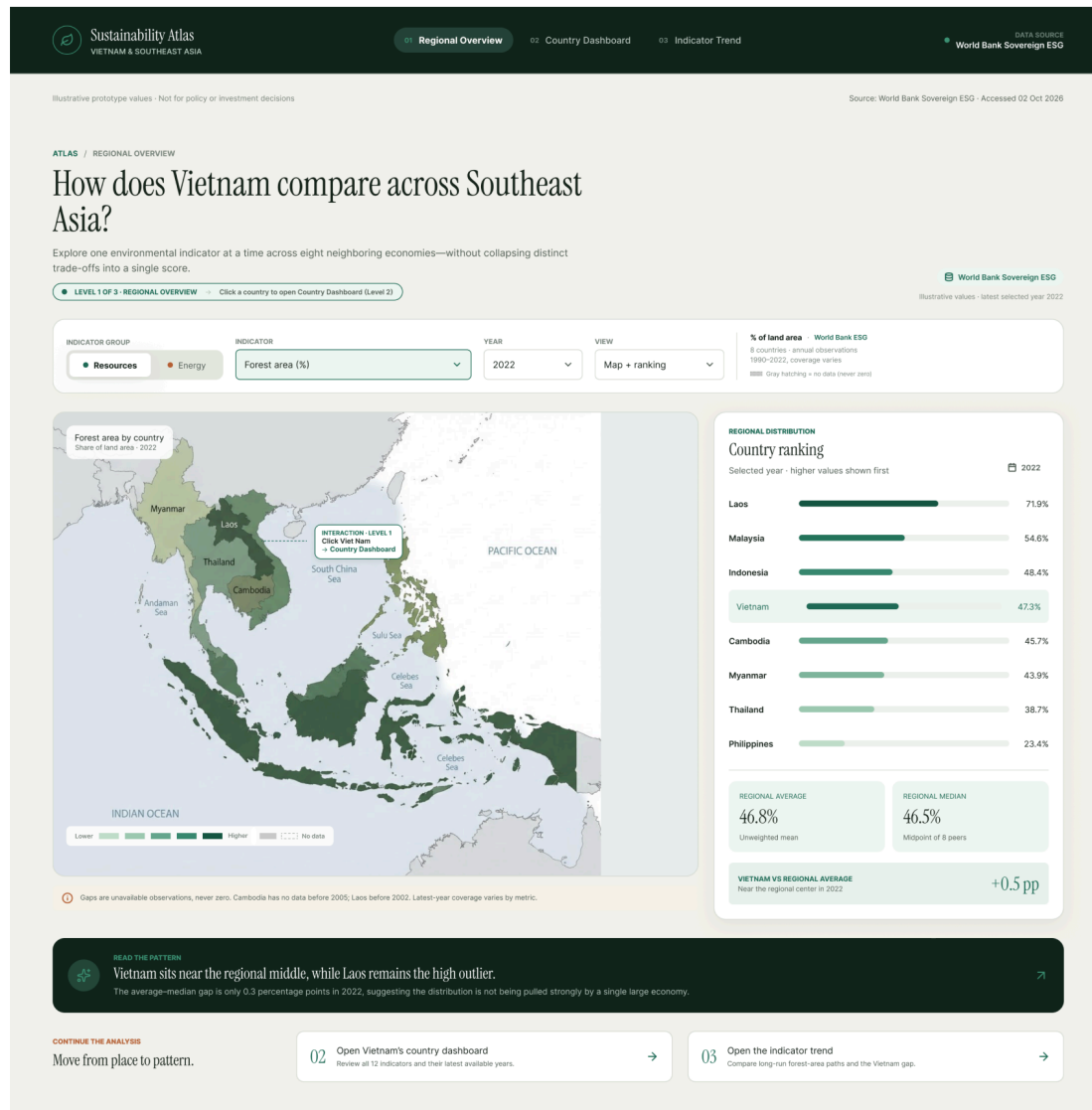
2021

▲ better

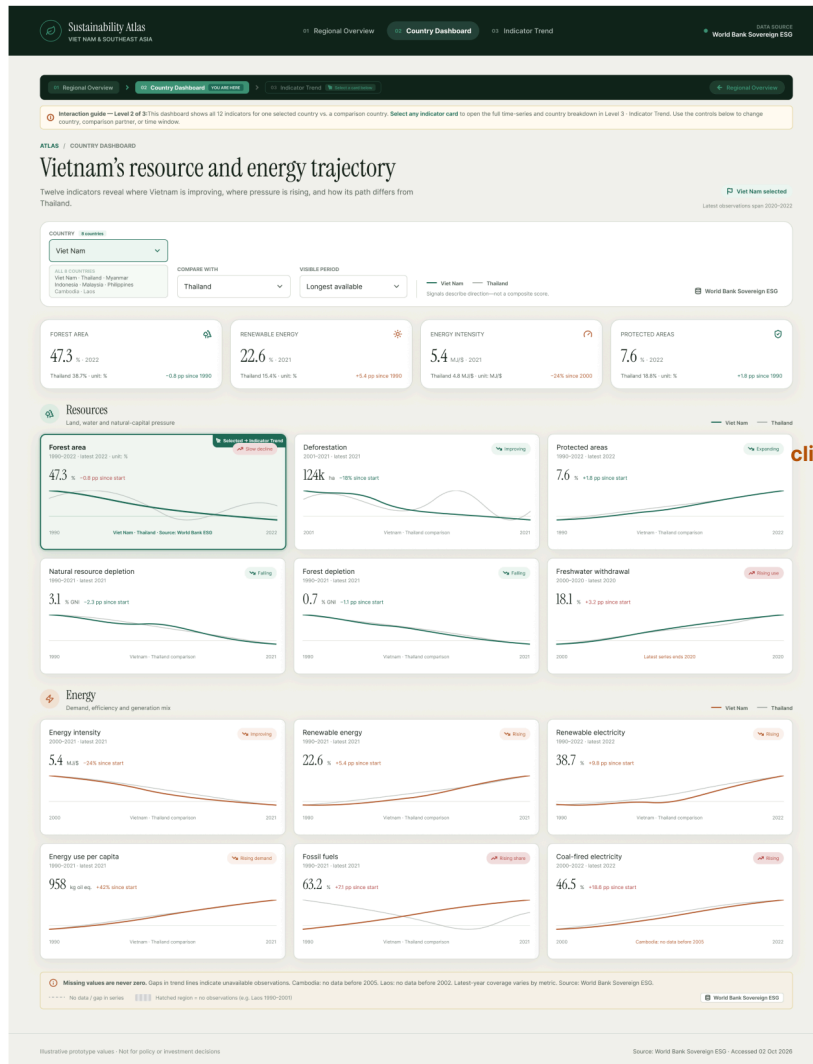
Regional reference is the MEDIAN of the 8 countries, never the mean. Missing values are shown as "no data", never 0. Ranks are direction-adjusted, so rank 1 is always the best outcome.

Source: World Bank Sovereign ESG · esgdata.worldbank.org

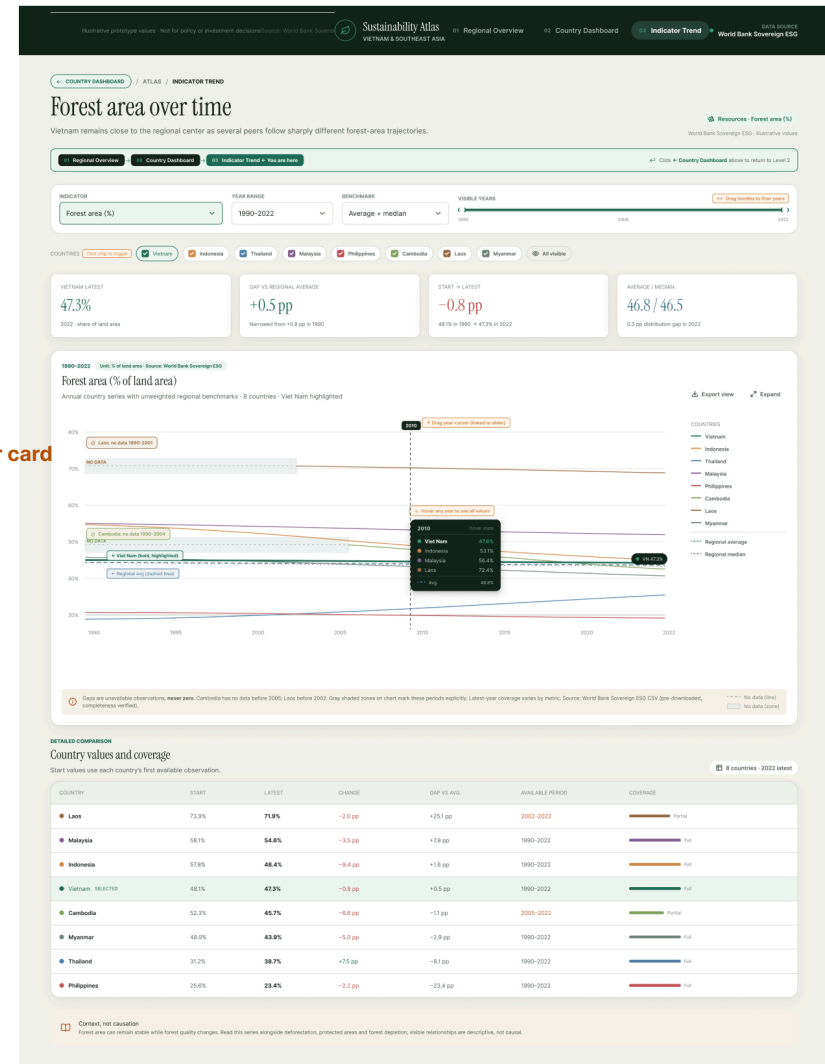
Sketch 4 — Design B: one click updates the KPI strip, small multiples and trend together.



Sketch 5 — Design C: three numbered levels. Level 1 ranks all eight countries and names the next step.



click an indicator card



Sketch 6 — Design C: clicking an indicator card opens its full trend; gaps are named, never drawn as zero.